The well:
five faces of the ground state of the windowed Weil form, and the approximation lemma that would hold them

D. Joubert (with the assistance of two language models)

6 September 2026 — consolidates notebook §16, §66–68, §86–89, §120–127 and Grok's `lemma-theta-v-consolidated.md` of `riemann-weil-phenomenology`

Abstract

On the window $[0, L]$, $L = \log\mu$, the truncated Weil form Q_L (archimedean part minus the prime towers up to μ) is positive with a smallest eigenvalue λ_0 that falls super-exponentially with μ — the *well*. Its ground state v has been measured from two sides over a week, by two independent chains (one on the space side, one on the zero side), on ζ, fifteen Dirichlet characters, eight elliptic curves and two Dedekind products, and the measurements fit one object: v is the unit-norm function of the window whose Fourier transform is hyper-null at the in-band zeros, whose spectral mass sits in the desert below the first zero, whose autocorrelation collapses super-exponentially at every lag beyond $y \approx 1.2$ (this is the "silence at the primes" of the quorum mechanism, which is not about primes), and whose value at the window edge is what remains: $\lambda_0 \approx \psi(0)^2 \cdot S$, with S the leakage of an edge jump onto the zeros beyond the band edge, computable from the zeros alone — the edge-doubling law $-\ln\lambda_0 = 2(-\ln|\psi(0)|) + O(1)$ found by Grok and verified on thirteen windows (the edge carries 82–103% of the depth). What is not established is the one thing that would turn the description into a depth law: how the minimal edge value depends on the configuration of in-band zeros. The desert alone does not fix it (dispersion by a factor 3 against $\tau\gamma_1$ on thirteen windows; the two-term formula and the one-set $\lambda_{\max}$ route are dead); the gap correction is $O(1)$ per cluster of zeros and its form is open. The note states the object, the evidence, the dead ends, and the approximation lemma. Nothing here bears on RH.

1 The object

Q_L on the hat/cosine basis $\{\eta_n\}_{n \le N}$ of $[0, L]$; $\lambda_0 = \min Q_L$, v its eigenvector, $\psi = \sum v_n \eta_n$ the function, $\hat{v}$ its Fourier transform, $\Theta_v(y) = \int \psi(t)\psi(t+y)\,dt$ its autocorrelation, $\ell = -\ln\lambda_0$ the depth, $s = \ell/\mu$ the drilling speed, $\omega_{\max} = 2\pi N/L$ the band edge. Under RH, $Q_L(f) = \sum_\gamma |\hat{f}(\gamma)|^2$ over the zeros; the repository's identity $Q_L =$ zero Gram holds to the tail on every L-function tried (`zeros-from-the-radical`, `gl2-prime-side`).

2 Five faces

1. The desert (§66–68, sampling-floor). λ_0 is the sampling constant of the zero set on the window, made small by the desert $[0, \gamma_1)$ where there are no zeros: under RH $c_L > 0$ (Theorem 1 of `sampling-floor`) and $\lambda_0(N) \downarrow c_L$. The Slepian cost of the desert alone, $1.71\, L(\gamma_1 - \nu)_+/\mu$ per unit μ, is 60–90% of s on the fast characters and on the rank-0 elliptic curves.

2. Hyper-nullity and the edge budget (§16.3–16.7). $|\hat{v}(\gamma_k)|$ at the in-band zeros is exponentially smaller than any bound requires (4.7×10^{-72} at γ_1 for ζ, $\mu = 16$) — the extremity law $|\hat{v}(\gamma_1)| \approx C\lambda_0$; the budget $\lambda_0 = \sum_k |\hat{v}(\gamma_k)|^2$ is carried by the zeros at and beyond the band edge (leakage rate $\tau \approx 0.48$, §16.4).

3. The mass in the desert (§122). $|\hat{v}(\gamma)|^2$ is 0.8 at $\gamma = 1$, 0.25 at 5, 5×10^{-3} at 10, then $e^{-1.34\gamma}$ with hyper-null dips at the zeros, 10^{-60} at $\omega_{\max}$ (ζ, $\mu = 16$): the norm lives below γ_1. Budget at the edge, mass in the desert — both true, distinct.

4. The collapse of the autocorrelation (§120–123; Grok's space-side lemma). $\Theta_v(y) > 0$ everywhere, smooth (no band-edge oscillation at step 0.004), and $-\ln \Theta_v(y)$ grows super-exponentially: $\approx 0.2\, s\, y\, e^y$ on ζ at $\mu = 16$ ($R^2 = 0.9995$, slope ratio $2.9 = (1+y)e^y$, Gaussian excluded), with a coefficient $\propto 1/L$ across μ ($a \cdot L = 0.62, 0.62, 0.58$ at $\mu = 8, 11, 16$) and a shape that the window edge bends: steeper near L, gentler for composite objects. The space side (Grok) reads the same: ψ has a Gaussian bulk, $\psi(t) \approx \psi_{\text{mid}} e^{-a(t-L/2)^2}$ with $aL^2 = \ell$ on ζ, and steep edges (e^{-s}); the $y\, e^y$ tail of Θ_v is the edge, not the desert weight. The towers sample this collapse at the prime lags: $-\ln \delta_p \approx 0.19\, s\, p \log p$ (one $\Gamma_{\mathbb{R}}$) or $0.38\, s\, p \log p$ (several; degrees 2 and 3 alike, §118–119) is the same law read at $n = p^k$, and at lags with no prime the same numbers appear (§120–121). Consequence for the quorum mechanism: its first half (the ground state is deaf to each prime) is a generic property of the well; only the coupling κ_p, Gaussian in $(p \log p)^2/(\mu \log \mu)$, is arithmetic. The ground state's sum rule sees only 2 and 3 ($98.9\% + 1.1\%$ for ζ).

5. The edge value (Grok; §125, §127). $-\ln \lambda_0 = 2(-\ln |\psi(0)|) + R$, $\psi(0) = L^{-1/2}(v_0 + \sqrt{2}\sum_{n \geq 1} v_n)$: ratio 2.06–2.19 on ζ, χ_3, χ_{29}, $\zeta L(\chi_3)$; $-2\ln|\psi(0)|/\ell \in [0.82, 0.98]$ on Grok's eleven windows and 0.97–1.03 on ζ — thirteen windows, five orders of magnitude in λ_0, the edge carries the depth. And why: a function with value $\psi(0)$ at both ends has a jump whose transform is $2\psi(0)\sin(\gamma L/2)/\gamma$; in band the smooth part cancels it (hyper-nullity), beyond the band it does not. On ζ at $\mu = 11$ with 500 zeros, the zeros beyond $\omega_{\max}$ carry 93.3% of λ_0 and the estimated tail beyond 811 completes it to 1.5%; the jump prediction $\sum_{\gamma_k > \omega_{\max}} 8\psi(0)^2 \sin^2(\gamma_k L/2)/\gamma_k^2$ matches to a factor 1.55 in the sum, predicting $R = 3.37$ against 3.29 measured. Hence

$$\lambda_0 \;\approx\; \psi(0)^2 \cdot S, \qquad S = \sum_{\gamma_k > \omega_{\max}} \frac{8\sin^2(\gamma_k L/2)}{\gamma_k^2} \times O(1.5) \approx \frac{4}{\pi}\,\frac{\log(\omega_{\max}/2\pi e)}{\omega_{\max}},$$

which explains why R depends on N (through $\omega_{\max}$) and on the object (through the zero density), and why it changes sign when in-band zeros near the band edge still carry budget (ζ, $\mu = 16$: $R = -4.4$).

3 One object

The five faces are one statement: *the ground state is the unit-norm function of the window that is hyper-null at the in-band zeros and minimizes its value at the edge; its mass sits in the desert because that is where no constraint holds; its autocorrelation collapses because a smooth bump with steep edges has one; and the depth is the square of the minimal edge value times the leakage of the edge jump onto the zeros beyond the band.* The primes enter twice, and only twice: as the samples of the collapse (harmless), and in the coupling κ_p that interlocks the towers (the arithmetic of the quorum).

4 What is dead

The two-term formula $-\ln c_L \approx aL(\gamma_1 - \nu)_+ + bL\sum(\text{gap} - \nu)_+$ (its gap sum does not converge; at a common cutoff it overpredicts most L-functions, §92–94). The one-set constant by $\lambda_{\max}(\chi_E P \chi_E)$ ($\lambda_{\max}$ sees only the largest interval; Grok, `lemma2-sampling-constant`). The line $\ell \approx a\tau\gamma_1 + b$ (jackknife: the slope halves without the one deep window). The Gaussian form of the collapse in y (slopes). The reading of the silence as arithmetic (§120). Ten pre-registered executions died on this object in one week; the object survived all of them.

5 What is open: the approximation lemma

Lemma 1 (to prove)**.** *Let* $\Gamma_L = \{\gamma_k < \omega_{\max}\}$ *be the in-band zeros. Among unit-norm functions on* $[0, L]$ *whose Fourier transform vanishes on* Γ_L *(to the hyper-null order observed), the minimal edge value satisfies* $-\ln|\psi(0)|_{\min} = \frac{1}{2}\, s\, \mu + O(1)$*, i.e.* $\lambda_0 \asymp \psi(0)^2_{\min}\, S(\omega_{\max})$*, with* s *a functional of the configuration* Γ_L*:* $s = C_0\, \tau\gamma_1/\mu +$*(gap correction), the correction* $O(1)$ *per cluster of zeros.*

What the data say about s: the desert alone gives a factor-3 dispersion ($\ell/\tau\gamma_1 \in [2.3, 7.4]$ on thirteen windows, ζ the deepest per unit of desert); the gap correction is $O(1)$ per piece in the log-determinant of the one-set compression (Widom's perimeter term: 0.46–0.62 nats per gap, position-independent), which matches ℓ to 20% on χ_5 only; no elementary (γ_1, L) predictor of $|\psi(0)|$ exists. The lemma is pure harmonic analysis — an extremal problem with linear constraints on a window — and it contains no prime. Its proof would give the depth law; its failure would tell which constraint the picture misses.

Status

Every number above is measured, with its script and its judge in the repository; the identity $Q_L =$ Gram is the only theorem used and it is used under RH. The two chains — space side (Grok: ψ, $aL^2 = \ell$, edge doubling) and zero side (this note: budget, mass, collapse, leakage) — were run independently and agree. Nothing here bears on RH.